Paroxysmal Sympathetic Hyperactivity (PSH) and Agitation after Acquired Brain Injury

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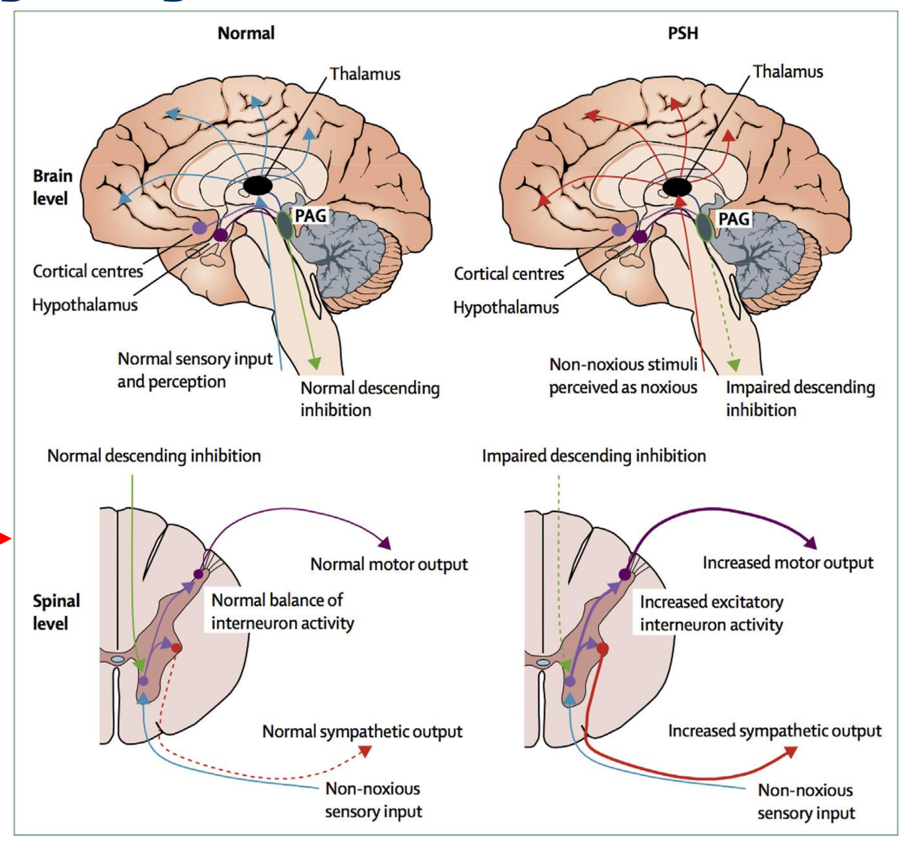
After the acute phase of brain injury patients may experience either paroxysmal sympathetic hyperactivity (sympathetic storming) or agitation. Agitation is seen in various neurological conditions, while sympathetic storming is seen predominantly after severe traumatic brain injury due to autonomic hyperreflexia. Both make weaning from sedation and mechanical ventilation difficult.



From Meyfroidt G, Baguley IJ, Menon DK. Paroxysmal sympathetic hyperactivity: the storm after acute brain injury. Lancet Neurol 2017;16(9):724

SYMPATHETIC STORMING

GCS ≤ 8

Episodic and exaggerated response occurring spontaneously or to stimulus

Resolves over time but may take weeks

Should be managed in short term as symptoms can be severe and may exacerbate brain injury

Symptoms

- Tachycardia
- Tachypnoea
- Diaphoresis
- Hypertension
- Abnormal movement of limbs
- Extensor posturing

Treatment

(Best practice based)

Atenolol NG
or **Clonidine** infusion
(20-80mcg/h)
(600mcg in 30ml 0.9% saline – concentration 20mcg/ml)
(see clonidine protocol)

Titrate until symptom control *(Observe for bradycardia and hypotension)*

Consider weaning clonidine after 24-48 hours free of 'storming' – stopping abruptly may cause 'hypertensive crisis' especially if patient has been on large dose or for >1 week

Avoid over-stimulation

May require individual patient parameters to avoid over-treatment with sedation

Early mobilisation / tilt table

MONITOR FOR DRUG SIDE-EFFECTS

Cardiovascular:
QT prolongation
Hypotension

Neurological:
Extrapyramidal symptoms

AGITATION

GCS >8

Disturbed behaviour seen as an early symptom in patients with post-traumatic amnesia (PTA)

Resolves with return of ability to retrieve and store information

In longer term it is usually due to personality changes

Symptoms

- Confusion
- Restlessness / agitation
- Limited awareness
- Disturbed perception of environment
- Inability to learn new tasks

Treatment

(Best practice based)

Rapid control of acute agitation
Lorazepam / **Haloperidol**
see algorithm

To facilitate weaning consider the following prior to stopping sedation:

NG **Risperidone** 0.5-2mg/tds or **Clonidine** infusion 20-80mcg/h
(see clonidine protocol)

Introduce NG clonidine [0.1mg tablets] if needed for longer-term treatment

Avoid using sedation / analgesia to treat agitation *(but exclude pain as cause)*

Exclude other causes - *UTI / metabolic disturbance*

Early mobilisation

Minimise sensory stimulation

Promote normal sleep pattern

ALCOHOL WITHDRAWAL RELATED AGITATION

Sweating, **palpitations** and **tremors** in association with agitation may be seen in patients withdrawing from alcohol – if history of alcohol excess, start **Chlordiazepoxide**

RAPID CONTROL OF ACUTE AGITATION

1
Implement behavioural interventions, structure environment and minimise sensory stimulation*

RESPONSE →

2
Consider adding oral medication e.g. Risperidone 0.5 - 4mg/day
OR
Carbamazepine 400 - 600mg BD
Tail off over days or weeks

NO RESPONSE ↓

3
Give either:
IV or IM Lorazepam 1-2 mg
Wait 10 minutes
or
IM Haloperidol 2-10mg
or
IV Haloperidol 2.5-5mg
Wait 30 minutes

RESPONSE →

4
Add oral medication as at ***2*** above

NO RESPONSE ↓

5
Repeat 3 to maximum:
IM Haloperidol 18mg /
IV Haloperidol 10mg
or
Lorazepam 4mg
Consider Neuropsychiatry referral

RESPONSE → 4

NO RESPONSE ↓

6
Seek NCC Consultant advice

Dexmedetomidine
See protocol

Only used after standard sedative agents trialled for 48 hours
(i.e. propofol, clonidine or benzodiazepine)

Consider in actual or suspected hyperactive delirium where agitation precludes weaning from sedation and extubation

***Environmental factors**

Noise reduction
Day - ↑ exposure to natural light
Night - ↓ exposure to artificial light
Optimum ambient temperature
Improved communication
Repeat orientation to day, time, place

- Guidelines are largely best practice based
- Exclude other causes of agitation e.g. urinary retention
- Give IV/IM drugs over 2-3 min - *never give Diazepam IM*
- In elderly, doses of 50% or less may be appropriate
- Guidelines do not apply to later maladaptive behaviour after brain injury